

CYBERSECURITY | WINDOWS ADMINISTRATION | ACTIVE DIRECTORY

Windows Active Directory & Group Policy Home Lab

Building and Managing a Windows Domain Environment

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About the Project

This is a hands-on Windows infrastructure and cybersecurity home lab. I built it to simulate a small organization with a central domain, departments, users and managed workstations.

The goal was practical experience managing a centralized Windows environment, not treating every PC as a separate island. The domain in my lab is test.local.

Everything in this deck comes from my own lab notes and screenshots. Where something was not documented, I say so.

Windows Server 2025

Windows 11 Pro

Active Directory DS

DNS

Organizational Units

Users

Security Groups

Group Policy

Logon / startup scripts

Password policy

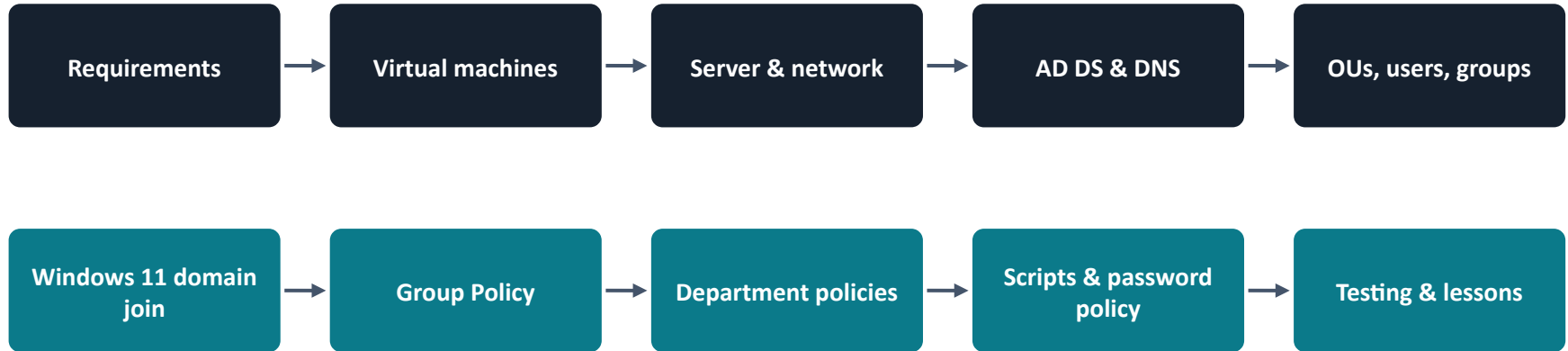
Domain-joined client

What I Wanted to Learn

- How Active Directory works
- What a Domain Controller does
- Creating and managing users
- Organizing users with OUs
- Creating security groups
- Joining computers to a domain
- Why DNS matters for Active Directory
- Creating and managing Group Policies
- Applying policies per department
- Managing Windows clients centrally
- Running scripts through Group Policy
- Applying password security controls
- Troubleshooting a Windows domain

Project Journey

The deck follows the same order I built the lab in.



Machine setup > Directory services > Policy management > Verification

Lab Requirements

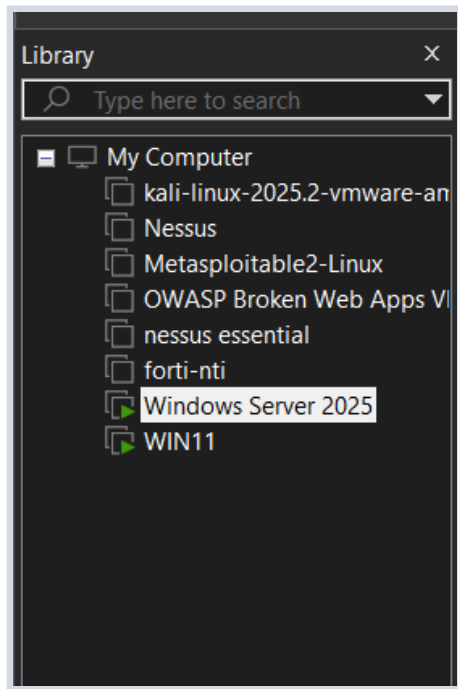
1. Windows Server 2025

Role: Domain Controller. Hosts Active Directory Domain Services, DNS and Group Policy management. Computer name in my lab: ADMIN-Server.

2. Windows 11 Pro

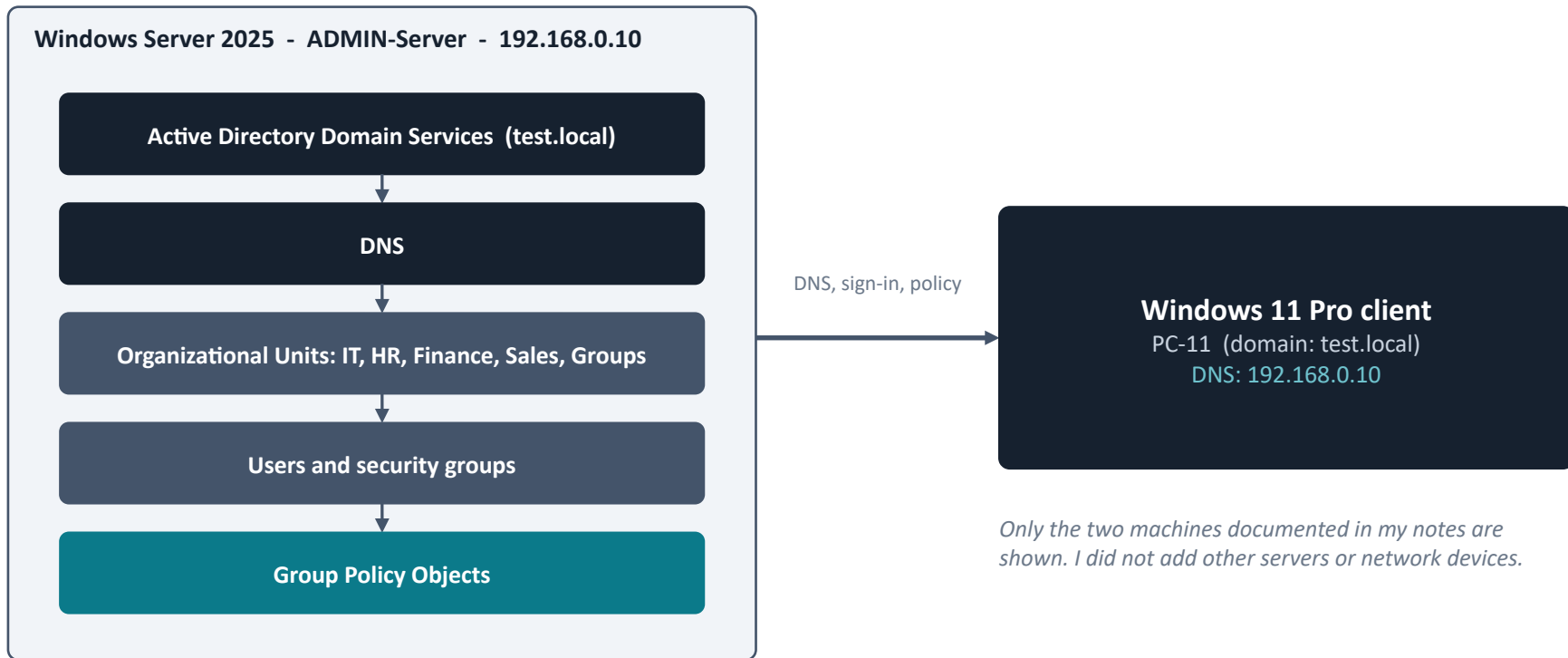
Role: domain-joined client workstation used to receive users and policies. Additional Windows 11 machines could represent more users or departments.

Both machines run as virtual machines. The VM library on the right shows the two lab machines.



VM library: Windows Server 2025 and WIN11

Lab Architecture



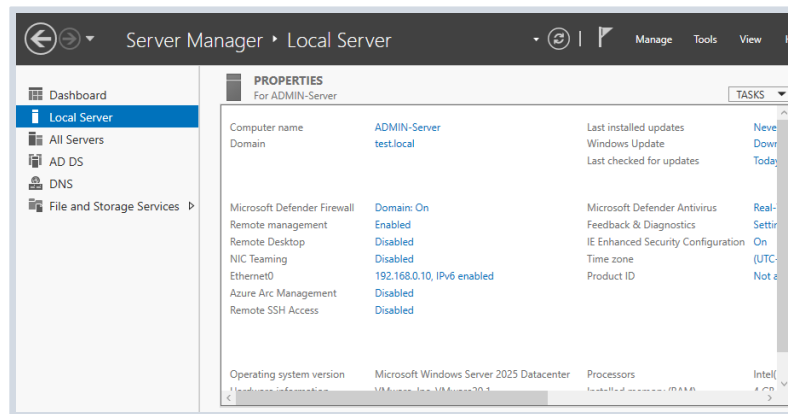
Windows Server Preparation

WHAT Installed Windows Server 2025 and gave the server the static IP 192.168.0.10.

HOW Set the Ethernet IP from the network adapter settings (ncpa.cpl is the tool in my command list). Server Manager then shows Ethernet0 as 192.168.0.10.

WHY A Domain Controller needs a predictable address. Clients and other services rely on it for DNS and for finding Active Directory, so it must not change.

RESULT A fixed, reachable server ready for the AD DS role. The server shows as ADMIN-Server in domain test.local.



Server Manager > Local Server (ADMIN-Server, test.local, Ethernet0 192.168.0.10, Windows Server 2025 Datacenter)

NOTE Subnet mask, gateway and server date/time steps are not explicitly documented in the original lab notes.

Windows 11 Client Preparation

WHAT Prepared the Windows 11 Pro machine before joining it to the domain.

HOW Installed Windows 11 Pro, changed the PC name, set the date and time, then assigned an IP in the 192.168.0.1X range with DNS 192.168.0.10.

WHY The client finds the domain through DNS. Pointing DNS at the Domain Controller lets it resolve test.local and locate Active Directory services.

RESULT The client can resolve the domain and is ready to join it.

Setting	Value in my lab
Operating system	Windows 11 Pro
Computer name	PC-11 (notes say PC11)
Client IP	192.168.0.1X range
DNS server	192.168.0.10
Date & time	Changed (value not recorded)

The exact client address and the PC11 vs PC-11 spelling are shown as documented: my notes say PC11, the screenshot shows PC-11.

Installing Active Directory Domain Services

What it is and why it matters

Active Directory Domain Services (AD DS) is the Windows directory service. It stores users, groups and computers in one place and handles authentication for the domain.

In an enterprise it replaces per-PC accounts with central identity and policy. That is why my project needs a Domain Controller: without one there is nothing to join, sign in to, or apply policy from.

WHAT Installed AD DS on the server and created the domain.

HOW Per my notes: set up Active Directory and assigned 192.168.0.10. Server Manager lists AD DS and DNS, and the domain is test.local.

WHY To get a central directory that can hold OUs, users, groups and GPOs.

RESULT Domain test.local running on ADMIN-Server.

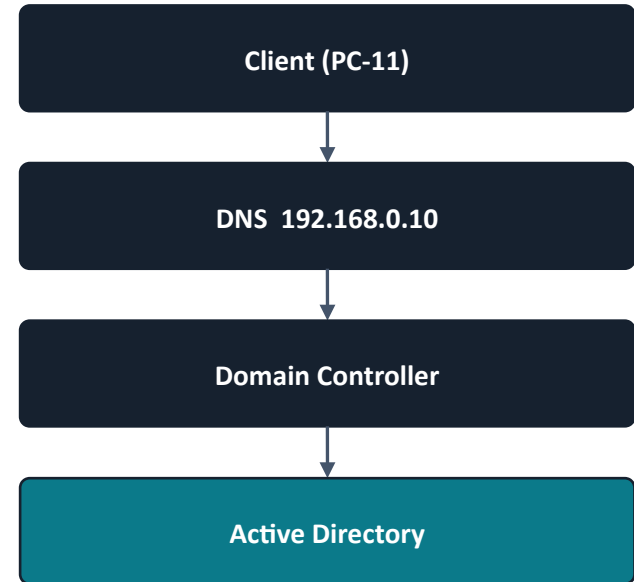
NOTE Promotion wizard screens and functional levels are not explicitly documented in the original lab notes.

Domain Controller Configuration

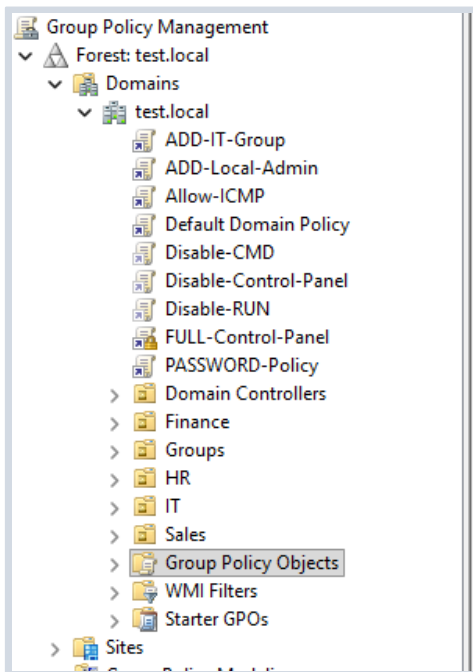
Item	Value (from my screenshots)
Operating system	Windows Server 2025 Datacenter
Computer name	ADMIN-Server
Domain	test.local
IP address	192.168.0.10 (Ethernet0)
Roles shown	AD DS, DNS, File and Storage Services
Platform	VMware virtual machine, 4 GB RAM visible

The client asks DNS where the domain is. DNS on the Domain Controller answers, and the client then talks to Active Directory.

How a client reaches the directory



Organizational Units



GPMC tree: Finance, Groups, HR, IT, Sales

WHAT Created OUs for the departments: IT, HR, Finance and Sales, plus a Groups OU for the security groups.

HOW In Active Directory Users and Computers (dsa.msc).

WHY An OU is a logical container for users and computers. Departmental OUs keep the directory tidy, let me apply different GPOs per department, and allow delegated administration.

RESULT Each department can be managed and policed separately. HR, Sales and Finance each carry their own GPOs.

Creating Users

WHAT Created users for the departments (IT, HR, Finance, Sales) and an Admin group.

HOW Users are created in Active Directory Users and Computers (dsa.msc) inside the department OU.

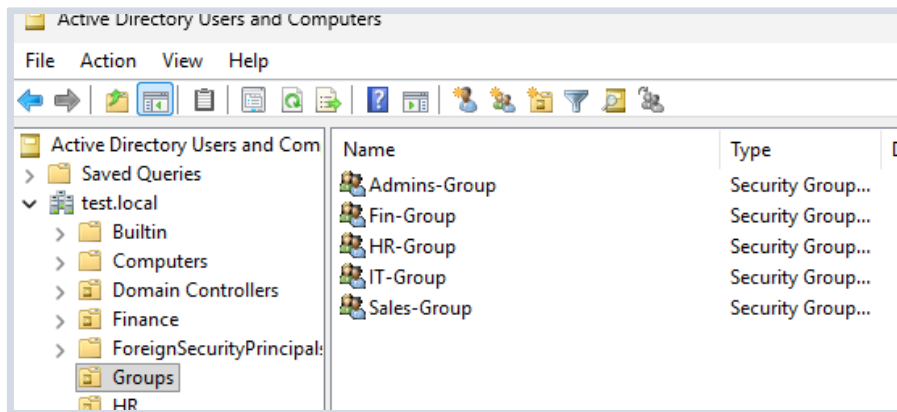
WHY Separate departmental users let me test different policies on different people, and central accounts mean one place to manage sign-ins.

RESULT Domain users that can sign in on domain-joined PCs and receive the policies of their OU.

NOTE My screenshots do not show individual user account names, so I do not list any. Usernames and passwords are not recorded in the original notes.

Centralized user management means one account works across domain PCs and one change applies everywhere.

Security Groups



Active Directory Users and Computers: Groups OU

Group	Type	Matches
IT-Group	Security Group	IT
HR-Group	Security Group	HR
Fin-Group	Security Group	Finance
Sales-Group	Security Group	Sales
Admins-Group	Security Group	Admin

Why groups: assign access to a group instead of to each person.

With GPOs: groups can be used for filtering and for local admin rights (IT-Group is used that way).

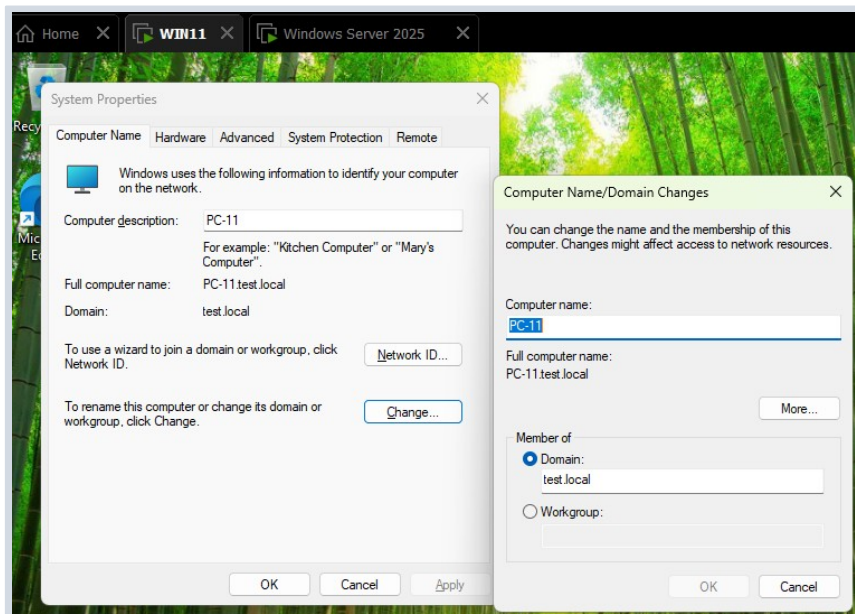
Matches column: inferred from the group names.

Joining Windows 11 to the Domain

- 1 Configure the client IP
- 2 Set DNS to 192.168.0.10
- 3 Confirm DNS points to the Domain Controller
- 4 Change the computer name
- 5 Open System Properties (sysdm.cpl)
- 6 Choose Change > Domain: test.local
- 7 Enter domain credentials
- 8 Restart the computer
- 9 Sign in with a domain account
- 10 Check the computer in Active Directory

NOTE Teal steps are backed by my notes or screenshots. Steps 7 to 9 are the standard process and were not captured as screenshots.

Domain Join in Practice



Windows 11: System Properties > Change > Domain: test.local

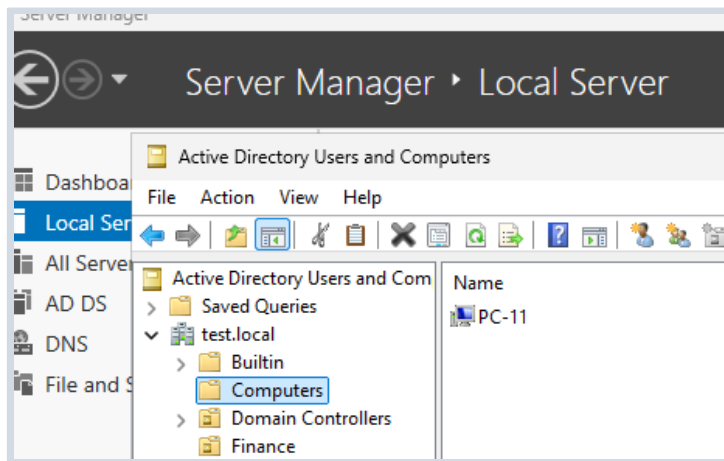
WHAT Joined the Windows 11 client to test.local.

HOW Set the DNS server to 192.168.0.10 first. Then in System Properties > Change, selected Domain and entered test.local. The screenshot shows the full name PC-11.test.local.

WHY Domain join looks up the domain through DNS. If DNS points elsewhere, the join fails.

RESULT The client became a member of test.local and could be managed by Group Policy.

Domain Environment Verification



PC-11 listed in the Computers container

Joined: System Properties shows the domain test.local and full name PC-11.test.local.

Computer object: PC-11 appears in the Computers container in dsa.msc (screenshot).

Domain users sign in: sign in as a domain account on the client.

Policies reach the client: run gpupdate /force, then check the expected behavior of each GPO.

The first two items are shown in my screenshots. The last two are the checks I used in testing.

Introduction to Group Policy

Group Policy lets an administrator centrally control settings for users and computers in a domain. Instead of changing each PC, I change a GPO once and it applies everywhere it is linked.

User Configuration

Settings that follow the person who signs in: desktop, Start menu, Control Panel and Task Manager access, and similar restrictions.

Many of my department restrictions belong here.

Computer Configuration

Settings that apply to the machine no matter who signs in: security settings, account policies, scripts at startup.

My PASSWORD-Policy is edited here (Computer Configuration > Windows Settings > Security Settings > Account Policies).

Group Policy Structure

Domain test.local - GPOs linked at the domain

Default Domain Policy | FULL-Control-Panel (enforced) | ADD-Local-Admin | ADD-IT-Group | Allow-ICMP | PASSWORD-Policy



OU: HR

Remove-Clock, Disable-Properties, Disable-USB, Enable-Conrol_panel, Disable-Task-Manager, Creat-ShourtCut

OU: Sales

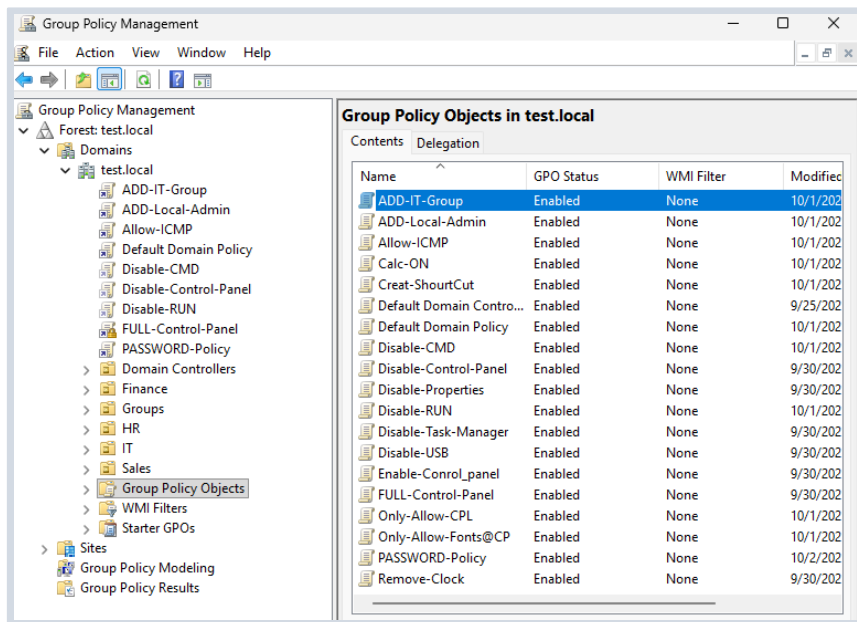
Disable-USB, Disable-Task-Manager, Disable-Properties, Remove-Clock, Calc-ON

OU: Finance

Calc-ON, Disable-Properties, Disable-USB, Remove-Clock, Disable-Task-Manager

Linked GPOs per OU are read from the Group Policy Inheritance tab in my screenshots. I have no policy screenshot for the IT OU, so I do not claim any.

Group Policy Objects in the Domain

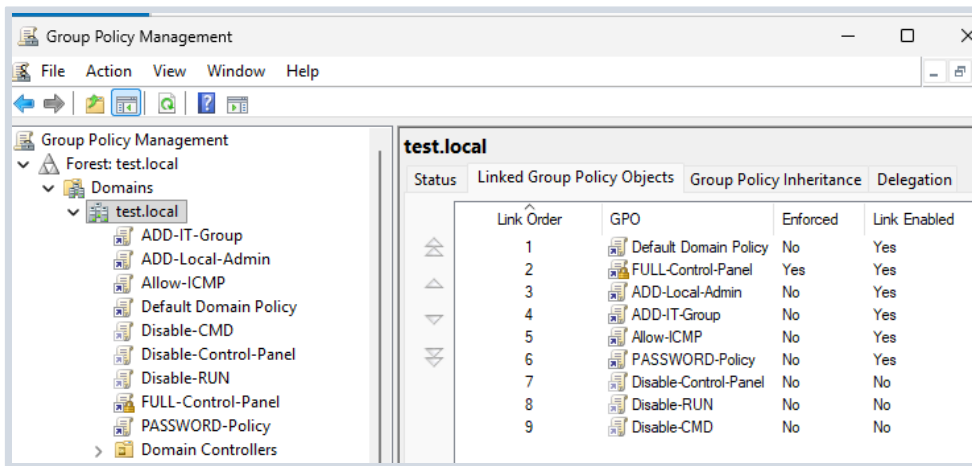


Group Policy Objects in test.local: 19 GPOs, all Enabled

Group	GPO names (as shown in console)
Restrictions	Disable-CMD, Disable-Control-Panel, Disable-Properties, Disable-RUN, Disable-Task-Manager, Disable-USB
Control Panel	Enable-Conrol_panel, FULL-Control-Panel, Only-Allow-CPL, Only-Allow-Fonts@CP
Desktop	Remove-Clock, Creat-ShourtCut, Calc-ON
Access	ADD-IT-Group, ADD-Local-Admin, Allow-ICMP
Security	PASSWORD-Policy
Defaults	Default Domain Policy, Default Domain Controllers Policy

Names keep the exact spelling from the console, including my typos (Conrol, ShourtCut).

All Users Group Policy (test.local)



Linked Group Policy Objects on test.local

Order	GPO	Enforced	Link
1	Default Domain Policy	No	Yes
2	FULL-Control-Panel	Yes	Yes
3	ADD-Local-Admin	No	Yes
4	ADD-IT-Group	No	Yes
5	Allow-ICMP	No	Yes
6	PASSWORD-Policy	No	Yes
7	Disable-Control-Panel	No	No
8	Disable-RUN	No	No
9	Disable-CMD	No	No

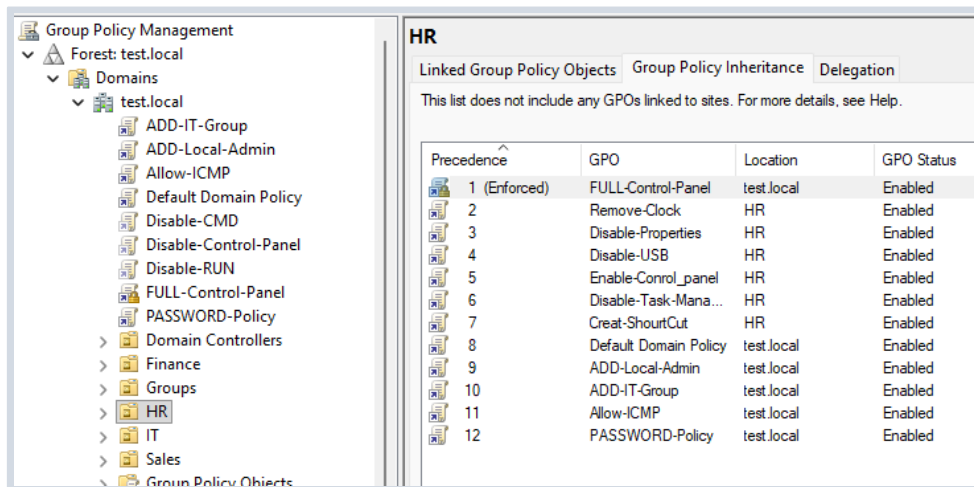
Scope: every user and computer in test.local. Policies shown with Link = No exist but are not applied at the domain level.

All Users: What Each Policy Does

GPO	What it does (by name / my notes)	What users see	How to test
FULL-Control-Panel (enforced)	Control Panel access rules; enforced so department GPOs cannot override it	Control Panel behaves per the policy on every OU	Open Control Panel as a user from each OU
ADD-Local-Admin	Runs the net user / net localgroup script to create a local admin	A local admin account exists on the PC	Check local users on the client after gpupdate /force
ADD-IT-Group	Adds IT-Group to local Administrators	IT-Group members have admin rights on the PC	Sign in as an IT-Group member and test admin tasks
Allow-ICMP	Allows ICMP (ping) through the client firewall	The PC answers ping	Ping the client from the server
PASSWORD-Policy	Domain password and lockout rules	Password rules apply at next change; lockout after failures	See the password policy slides

Exact setting contents are not visible in the screenshots. I describe each GPO from its name and from the commands in my notes.

HR Group Policy



HR OU: Group Policy Inheritance tab

Precedence 1 is FULL-Control-Panel (enforced, from test.local). HR GPOs are 2 to 7. Domain GPOs follow (8 to 12).

POLICY HR OU GPO set

PURPOSE Standard desktop restrictions for HR, plus a shortcut

CONFIGURATION 6 GPOs linked to the OU (next slide)

SCOPE Users and computers in the HR OU, plus inherited domain GPOs

RESULT Restricted desktop as listed next

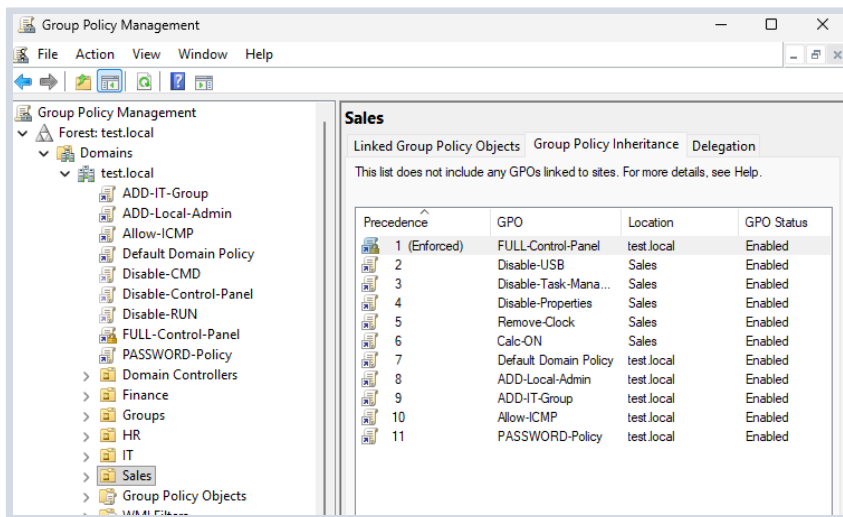
VERIFY Sign in as a HR user, run gpupdate /force, test each item

HR: Policy Details

GPO linked to HR	Purpose	Expected result	Verification
Remove-Clock	Removes the clock from the taskbar (by name)	Clock is gone from the taskbar	Check the taskbar after sign-in
Disable-Properties	Restricts Properties dialogs (by name; exact setting not visible)	Properties access is limited	Right-click an item and try Properties
Disable-USB	Blocks USB storage (by name)	USB drive cannot be used	Plug a USB drive into the client
Enable-Control_panel	Control Panel access for HR (by name; exact setting not visible)	Control Panel opens	Open Control Panel as an HR user
Disable-Task-Manager	Blocks Task Manager (name truncated in console)	Task Manager is blocked	Press Ctrl+Shift+Esc
Creat-ShourtCut	Creates a shortcut (by name; target not documented)	A shortcut is present at sign-in	Check the desktop after sign-in

Inherited from test.local: FULL-Control-Panel (enforced), Default Domain Policy, ADD-Local-Admin, ADD-IT-Group, Allow-ICMP, PASSWORD-Policy.

Sales Group Policy



Sales OU: Group Policy Inheritance tab

Precedence 1 is FULL-Control-Panel (enforced). Sales GPOs are 2 to 6. Domain GPOs follow (7 to 11).

POLICY Sales OU GPO set

PURPOSE Locked-down Sales desktop with Calculator auto-launch

CONFIGURATION 5 GPOs linked to the OU (next slide)

SCOPE Users and computers in the Sales OU, plus inherited domain GPOs

RESULT Restricted desktop as listed next

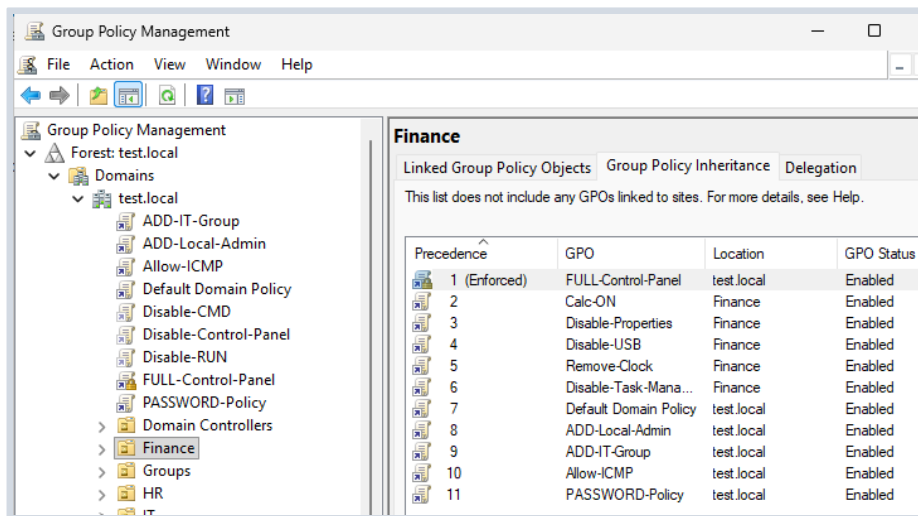
VERIFY Sign in as a Sales user, run gpupdate /force, test each item

Sales: Policy Details

GPO linked to Sales	Purpose	Expected result	Verification
Disable-USB	Blocks USB storage (by name)	USB drive cannot be used	Plug a USB drive into the client
Disable-Task-Manager	Blocks Task Manager (name truncated in console)	Task Manager is blocked	Press Ctrl+Shift+Esc
Disable-Properties	Restricts Properties dialogs (by name; exact setting not visible)	Properties access is limited	Right-click an item and try Properties
Remove-Clock	Removes the clock from the taskbar (by name)	Clock is gone from the taskbar	Check the taskbar after sign-in
Calc-ON	Auto-runs calc.exe (from my script notes)	Calculator opens at sign-in	Sign in and watch for Calculator

Inherited from test.local: FULL-Control-Panel (enforced), Default Domain Policy, ADD-Local-Admin, ADD-IT-Group, Allow-ICMP, PASSWORD-Policy.

Finance Group Policy



Finance OU: Group Policy Inheritance tab

Precedence 1 is FULL-Control-Panel (enforced). Finance GPOs are 2 to 6. Domain GPOs follow (7 to 11).

POLICY Finance OU GPO set

PURPOSE Locked-down Finance desktop with Calculator auto-launch

CONFIGURATION 5 GPOs linked to the OU (next slide)

SCOPE Users and computers in the Finance OU, plus inherited domain GPOs

RESULT Restricted desktop as listed next

VERIFY Sign in as a Finance user, run gpupdate /force, test each item

Finance: Policy Details

GPO linked to Finance	Purpose	Expected result	Verification
Calc-ON	Auto-runs calc.exe (from my script notes)	Calculator opens at sign-in	Sign in and watch for Calculator
Disable-Properties	Restricts Properties dialogs (by name; exact setting not visible)	Properties access is limited	Right-click an item and try Properties
Disable-USB	Blocks USB storage (by name)	USB drive cannot be used	Plug a USB drive into the client
Remove-Clock	Removes the clock from the taskbar (by name)	Clock is gone from the taskbar	Check the taskbar after sign-in
Disable-Task-Manager	Blocks Task Manager (name truncated in console)	Task Manager is blocked	Press Ctrl+Shift+Esc

Inherited from test.local: FULL-Control-Panel (enforced), Default Domain Policy, ADD-Local-Admin, ADD-IT-Group, Allow-ICMP, PASSWORD-Policy.

Testing Group Policies

- Sign in as a user from the affected OU
- Check whether the policy is applied
- Update Group Policy on the client
- Verify the expected behavior (USB, Task Manager, clock, Calculator)
- Compare behavior before and after the policy
- If it does not apply, check DNS, OU placement and link status

gpupdate

Refreshes policy that changed since the last refresh. Quick and normal for small edits.

gpupdate /force

Reapplies all policies, changed or not. I used it when a result looked wrong or I wanted a clean test.

Group Policy Scripts: Auto-run Calculator

Item	From my notes
Script	calc.exe
GPO	Calc-ON
Linked to	Sales and Finance OUs

WHAT Made Calculator open automatically for Sales and Finance users.

WHY It is a harmless way to prove that a GPO script really runs on the client.

RESULT Calculator launches when the script runs, which confirms scripts can automate actions.

Whether it ran as startup or logon was not explicitly documented in the original lab notes.

Creating a Local Admin Through Script

```
net user itadmin PASSWORD /add  
net localgroup Administrators itadmin /add
```

Command 1: creates a local user called itadmin (PASSWORD is the placeholder from my notes).

Command 2: adds itadmin to the local Administrators group.

Linked via: the ADD-Local-Admin GPO, at the domain level.

LAB ONLY A plaintext password in a script is a lab shortcut. Scripts in SYSVOL can be read by authenticated domain users, and a domain-wide local admin account is a real risk. Do not do this in production.

Adding IT-Group to Local Administrators

```
net localgroup administrators IT-Group /add
```

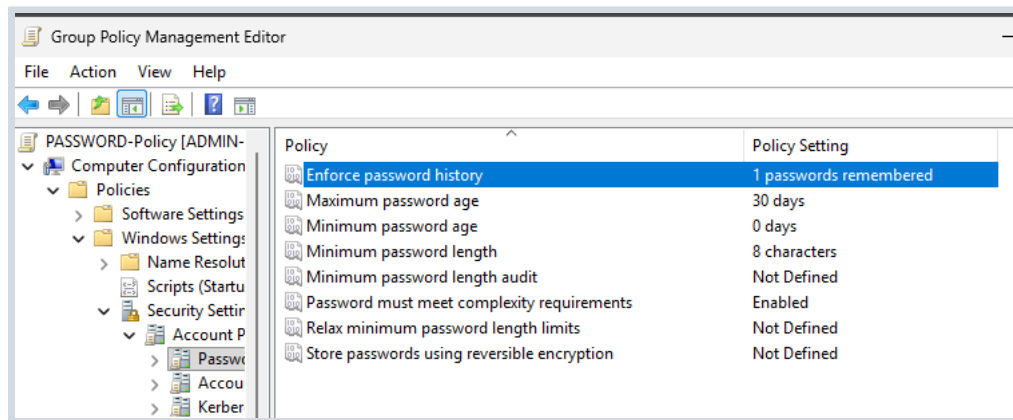
What it does: adds the domain group IT-Group to the local Administrators group on each client.

Why groups: IT staff are managed by group membership, so adding or removing a person changes their rights everywhere at once.

GPO: ADD-IT-Group, linked at test.local.

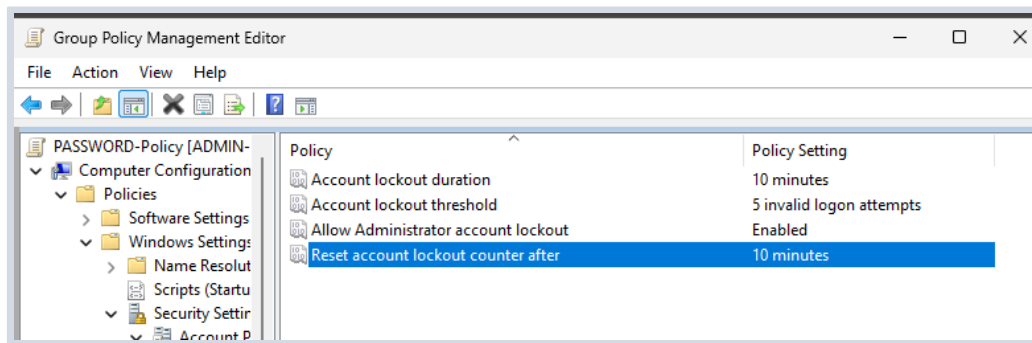
LAB ONLY This gives IT-Group admin rights on every domain PC in scope. In production I would limit it by least privilege, for example admin only on the machines that need it, and review membership regularly.

Password Policy: As Configured



Group Policy Management Editor, PASSWORD-Policy:
Computer Configuration > Windows Settings >
Security Settings > Account Policies.

Password rules and lockout rules were both set in the same GPO, PASSWORD-Policy, linked to test.local.



Password Policy: Values & Meaning

Setting	Value	What it controls
Enforce password history	1 password remembered	How many old passwords cannot be reused
Maximum password age	30 days	How long before a change is required
Minimum password age	0 days	How soon a user may change again
Minimum password length	8 characters	Shortest allowed password
Complexity requirements	Enabled	Mixed character types required
Account lockout threshold	5 invalid attempts	Failed logons before lockout
Lockout duration	10 minutes	How long the account stays locked
Reset lockout counter after	10 minutes	When the failure count resets
Allow Administrator lockout	Enabled	Admin account can also lock out

My review

History of 1 and a minimum age of 0 days make reuse easy, and a 30-day maximum age forces frequent changes. Today I would favor longer passwords and breached-password checks. Also check that Default Domain Policy (link order 1) does not override these values.

Windows Administrative Tools Used

Command	Purpose	Used for
ncpa.cpl	Network Connections	Change IP / DNS on the Ethernet adapter
sysdm.cpl	System Properties	Computer name and domain membership
dsa.msc	Active Directory Users and Computers	Manage users, groups, OUs, computers
gpedit.msc	Local Group Policy Editor	Local policy on one machine
gpmc.msc	Group Policy Management Console	Create and link domain GPOs
gpupdate	Update Group Policy	Apply policy that changed
gpupdate /force	Force policy refresh	Reapply all policies

My notes write the first command as cnpa.cpl. The correct Windows command is ncpa.cpl, and that is what this table uses.

Common Issues (1 of 2)

Problem	Possible cause	Check	Solution
Incorrect DNS	Client DNS is not the Domain Controller	ncpa.cpl: DNS should be 192.168.0.10	Correct the DNS server
Incorrect IP configuration	Wrong address or range	ncpa.cpl on server and client	Use a free 192.168.0.x address; server stays 192.168.0.10
Client cannot join domain	DNS or connectivity problem	Can the client reach 192.168.0.10?	Fix DNS first, then retry in sysdm.cpl
User cannot authenticate	Wrong credentials, lockout, or password rule	Account status in dsa.msc	Reset password or wait out the 10-minute lockout

Common troubleshooting considerations. These are generic checks, not issues I claim to have hit.

Common Issues (2 of 2)

Problem	Possible cause	Check	Solution
Group Policy not applying	Wrong OU, link disabled, policy not refreshed	gpmc.msc: link status and OU	Enable the link, move the object, run gpupdate /force
Computer not in Active Directory	Join did not finish, or wrong container	dsa.msc: Computers container	Rejoin via sysdm.cpl, then move to the right OU
Policy change not visible	Policy not yet refreshed	Wait for refresh or sign out	Run gpupdate or gpupdate /force
Incorrect OU placement	Object created in the wrong OU	dsa.msc: object location	Move the user or computer to the right OU

Common troubleshooting considerations. These are generic checks, not issues I claim to have hit.

Security Considerations

Centralized identity

One directory for all accounts

Least privilege

Admin rights only where needed

Group-based admin

Manage people by group

Password security

History, length, complexity, lockout

Policy enforcement

Enforced GPOs the OU cannot override

Domain management

Clients controlled from the server

Department separation

OUs with their own GPOs

Controlled admin access

IT-Group and itadmin scope

NOTE Some of my lab choices need hardening for production (next slide).

From Lab Shortcut to Production Practice

What I did in the lab	Production practice
Plaintext password in a script (itadmin)	Never store passwords in scripts; use a managed solution
ADD-Local-Admin linked to the whole domain	Control local admin creation; avoid domain-wide admin accounts
IT-Group added to local Administrators everywhere	Least privilege: admin only where needed, reviewed regularly
Policies applied straight to departments	Test GPOs on a pilot OU before broad deployment
History of 1, minimum age 0, 30-day expiry	Review against current guidance, favor length and breach checks

Skills Developed

Windows Server

Windows Server 2025, server configuration, network configuration

Active Directory

Domain Controllers, users, groups, OUs, domain management

DNS

DNS configuration, client DNS settings, DNS dependency for AD

Group Policy

GPO creation and organization, user and department policies, testing, updates

Windows Clients

Windows 11 configuration, domain joining, domain authentication

Automation

Windows commands, Group Policy scripts, basic administrative automation

What This Project Taught Me

Before this lab I thought of each Windows PC as its own machine. Building it showed me that in a domain, the PCs are managed from one place. I change a policy once and it follows the user.



Seeing a policy land on PC-11 after `gpupdate /force` made the idea concrete. I also saw how easy it is to over-grant access, which is why my security slides call out the weak spots.

Future Improvements

FUTURE ideas only. None of these are part of the current lab.

- Add more Windows clients
- Add a dedicated file server
- Implement more advanced GPOs
- Configure Windows auditing
- Enable centralized event logging
- Integrate a SIEM (Wazuh)
- Monitor Windows security events
- Create attack-and-defense scenarios
- Test common AD attacks in a controlled lab
- Implement backup and recovery scenarios
- Add PowerShell automation
- Add endpoint security monitoring

PROJECT SUMMARY

Built	A Windows domain (test.local) with a Domain Controller, a Windows 11 client, OUs, groups and department GPOs.
Used	Windows Server 2025, Windows 11 Pro, AD DS, DNS, Group Policy, scripts, password policy.
Learned	How central identity, groups, OUs and policy work together, and where lab shortcuts need hardening.
Why it matters	Domain management is the core of most enterprise environments I would defend or support.

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